# Weidong TANG 汤维栋

Incoming PhD Student at HKUST · CRO (Founding Member), Cardinal AI Lab, InfRec

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## EDUCATION

**Hong Kong University of Science and Technology** — Incoming
PhD, Department of Electronic and Computer Engineering
Advised by Prof. Song GUO, MAE, FCAE, FIEEE.

**Xidian University** — 09.2023 – 07.2027 (expected)
B.E., School of Telecommunication Engineering · GPA: 3.8/4.0
Advised by Prof. Xiumei WANG.

## EXPERIENCE

**Cardinal AI Lab, InfRec** — 07.2026 – Present
Chief Research Officer (Founding Member)
- Lead research on recursive self-improvement and long-horizon interaction, enabling agents to learn and control embodied robots.
- Guide research from problem formulation to prototypes, working with engineering teams on real-world embodied AI applications.

**Alibaba** — 05.2026 – 07.2026
MLLM Research Intern
- Research: combined multimodal representations with OPD, exploring post-training, SFT, and industrial product benchmarks.
- Business: studied image-text alignment data pipelines for product representations and retrieval in large-scale e-commerce.

**TeleAI** — 01.2026 – 03.2026
LLM Research Intern
- Focused on AI infrastructure for LLM training and inference on Huawei Ascend 910B, emphasizing efficient and stable execution.
- Worked on precision alignment for LLM training and inference, addressing precision differences at the AI infrastructure level.

**National University of Singapore** — 10.2024 – 10.2025
Research Assistant · HPC-AI Lab
Advised by Prof. Yang YOU, Prof. Wangbo ZHAO, and Dr. Pengfei ZHOU.
- Focused on efficient AI and multimodal model evaluation, studying model efficiency and performance across research tasks.
- Learned to lead research teams through collaborative projects, developing ideas and working toward research publications.

## PUBLICATIONS

* Equal contribution · † Corresponding author · ‡ Project leader

01 **The Trinity: Capability-Driven Recursive Self-Improvement for Multimodal Agents**
W. Tang*, R. Wang*, J. Zhang*, H. Wang, H. Chen, Z. Zhang, Y. Ma, Z. Jiang, Z. Guo, H. Li, Z. Jiao, Z. Cui, X. Yin, S. Liang, Z. Wan, H. Lin, W. Zhao, K. Zhang, P. Zhou†, O. Zhang†, & Y. You†.
arXiv preprint, 2026 · Tech Report · GitHub · Project Page · 机智流 · 智猩猩

02 **Route Anything: A Study of Model and Policy Routing for General-Purpose AI Systems**
P. Zhou, Y. Ma, H. Wang†, W. Tang, Z. Wan, W. Zhao, & Y. You†.
arXiv preprint, 2026 · Tech Report · GitHub · Project Page

03 **GroupToM-Bench: Benchmarking Group Theory of Mind and Nonlinear Social Emergence in MLLMs**
W. Tang, J. Li, Y. Hou, Z. Mei, C. Zhang, X. Wan, Z. Liang, P. Zhou†, Y. You, & W. Zhao†.
ACL 2026 (19% acceptance rate, CCF-A top-tier conference) · Paper · Dataset

04 **MagicBench: Diagnosing Visual Agency Loss and Semantic Dependency in Multimodal LLMs**
T. D. Huang, W. Tang‡, W. Q. Xu, & X. Guo†.
ACL 2026 (19% acceptance rate, CCF-A top-tier conference) · Paper · GitHub

05 **Efficient Video Object Segmentation and Tracking with Recurrent Dynamic Submodel**
**W. Tang**, Z. Liang, X. Wan, C. Zhu, Z. Xu, P. Zhou, Y. Song, Y. You, & W. Zhao†.
CVPR 2026 (25% acceptance rate, CCF-A top-tier conference) Paper

06 **Inference-Time Scaling for Visual AutoRegressive Modeling by Searching Representative Samples**
**W. Tang**, X. Wan, S. Li, & X. Wang†.
PRCV 2025 (29% acceptance rate, CCF-C) Paper · GitHub

07 **EA-ViT: Efficient Adaptation for Elastic Vision Transformer**
C. Zhu, W. Zhao†, H. Zhang, Y. Zhou, **W. Tang**, S. Wang, Z. Yuan, Y. Shang, X. Peng, K. Wang, & D. Yang†.
ICCV 2025 (25% acceptance rate, CCF-A top-tier conference) Paper · GitHub

08 **MDK12-Bench: A Multi-Discipline Benchmark for Evaluating Reasoning in Multimodal Large Language Models**
P. Zhou, X. Peng, F. Zhang, Z. Xu, J. Ai, Y. Qiu, W. Zhao, J. Song, C. Li, **W. Tang**, Z. Li, H. Zhang, Z. Li, X. Mao, Y. Feng, J. Sun, K. Wang, X. Chang, W. Shao, Y. You, & K. Zhang†.
AAAI 2026 (17% acceptance rate, CCF-A) Paper · GitHub

09 **IEBench: Benchmarking Unlearning Identity-Event Entangled Concepts for MLLM-Based Systems in Agentic Edge Networks**
Z. Xu, P. Zhou, **W. Tang**, J. Ai, W. Zhao, K. Wang, W. Shao, X. Peng, S. Zhao, H. Yao, & K. Zhang†.
IEEE Transactions on Network Science and Engineering, 2026 (IF=7.9, JCR-Q1) Paper

10 **Pebench: A Fictitious Dataset to Benchmark Machine Unlearning for Multimodal Large Language Models**
Z. Xu, P. Zhou, **W. Tang**, J. Ai, W. Zhao, K. Wang, X. Peng, W. Shao, H. Yao, & K. Zhang†.
arXiv preprint, 2025 Paper · Dataset · Project Page

11 **IndustryBench-MIPU: Benchmarking Multi-Image Attribute Value Extraction for Industrial Products**
H. Qi, J. Cao, Y. Zhang, X. Wang, **W. Tang**, B. Chen, C. Huo, H. Pan, H. You, J. Li, Y. Wang, & L. Ding.
arXiv preprint, 2026 Paper · GitHub · Dataset

12 **EarthVerse: Benchmarking Scientific Agents Across Dynamic Earth Systems and Natural Hazards**
Z. Cui, X. Yin, Y. Tang, X. Zhang, Y. Hu, S. Zhong, **W. Tang**, Y. Liang, W. Li, M. Jin, S. Pan, Y. Kang, D. Zhuang, & J. Zhao.
arXiv preprint, 2026 Paper · GitHub · Dataset · Project Page

## SELECTED PROJECTS

**WowPage**
A lightweight, open-source homepage template for researchers to showcase publications, projects, and academic profiles.

## TALKS

To be updated.

## REPORTS

**从科研博主到AI公司首席科学家：一名西电00后人工智能逐梦路** 10.2026
Interview by Qilu Evening News, a Shandong Provincial Media Outlet

**2026"创业之星"Next Star 一等奖** 11.2026
InfRec Wins First Prize in the Domestic Industry Semifinals and Advances to the Global Finals

**昌乐二中|271教育 汤维栋：纵有疾风起，人生不言弃** 11.2026
Outstanding Alumnus Interview

## AWARDS

BYD Scholarship 2026
Academic Scholarship, Xidian University (Top 5%) 2026
Academic Scholarship, Xidian University (Top 5%) 2025

## SERVICES

AI Team Lead, Xidian-Inspur Club 06.2025 – 05.2026
Coordinator, Xidian Grad Admissions Group (3,000 members) 10.2026 – Present